BUilders

A BURNING THREAT



- Gaps in community-wide wildfire plans leave neighborhoods vulnerable; Narrow roads in hilly terrains complicate evacuation and hinder fire engine access during emergencies
- Fire hydrants designed for small-scale fires, and during the Palisades Fire, water demand was four times higher than capacity; Elevated tanks ran dry within hours leaving firefighters without sufficient water
- Santa Ana Winds: gusts exceeding 60 mph fueled rapid fire spread by carrying embers to unburned areas and supplying fires with oxygen; Mountainous terrain and chaparral ecosystems spur wildfire spread and complicate implementation of wildfire-prevention infrastructure

PROBLEM STATEMENT

Our goal is to develop an AI-driven, autonomous wildfire prevention system to detect and resolve fires at the root, procuring safety within LA residents and promoting sustainability and equity.

AUDIENCE

Residents	City of L.A.	Community-Wide
Who: Homeowners of all income levels in wildfire-prone zones across LA **Pain Point**: high risk of property loss and slow emergency response due to location or lack of resources **FlameFreeze Helps**: Provides equal, automated protection adaptable to any home and budget	**Who**: Public agencies like LADWP, LAFD, and city planners managing fire and water infrastructure **Pain Point**: Strained firefighting resources and water systems during peak fire events **FlameFreeze Helps**: Eases the burden by distributing water independently and suppressing fires before they escalate. Incorporates preexisting infrastructures	**Who**: Groups of homes, HOAs, rural neighborhoods, and community centers vulnerable to fire spread **Pain Point**: Lack of coordinated, scalable protection systems for collective safety **FlameFreeze Helps**: Enables networked defense by syncing multiple properties under a shared AI-driven fire suppression system

SUSTAINABILITY & COST EFFICIENCY

GRAVITY-FED SYSTEM
Reduces energy costs by utilizing gravity for water distribution.

RECLAIMED STORMWATER
Uses stormwater instead of expensive fresh water.

SMART AUTOMATION
Reduces waste by spraying only where needed.

BIODEGRADABLE RETARDANTS
Prevents long-term environmental damage.

THE SOLUTION: AUTONOMOUS FIRE DETECTION & RESPONSE COORDINATION

FLAMEFREEZE
WILDFIRE PREVENTION

1

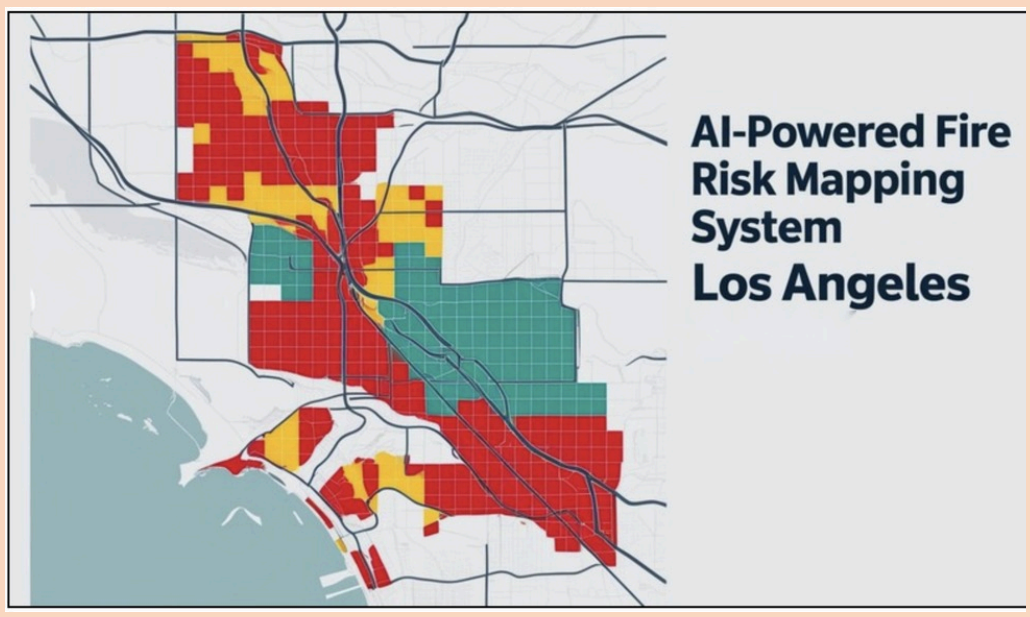
AI FIRE MAPPING

- Using Google's FireSat, detect fires and vulnerable areas

2

AUTONOMOUS WATER FLOW

- Underground water collection system from designated boreholes
- Water sent to vulnerable areas determined by AI



3

UPON FIRE DETECTION: DISPENSE WATER

- Dispense water using powerful sprinkler system on valley summits
- Rotating sprinkler heads adjust pressure as needed

4

DRONES COLLECT WEATHER DATA

- Define wind patterns, temperatures, and expected trajectories



5

DECAYING → **GROWING**

MONITOR STATUS

- Repeating steps 1-5 until fire no longer detected
- Ensure no relapse to growing fire

DISPENSE FIRE RETARDANTS

- Environmentally friendly gels/foams/aerosols dispensed from the powerful sprinkler system on valley summits to target areas

6

ALERT IMPACTED COMMUNITIES

- Assigned fire wardens to instruct residents on action plan: all resources, monitoring, and communication on our website with notifications through NotifyLA

SEND FIRE TRUCKS

- Manpower and firehoses to combat fire manually

LA SPECIFIC SOLUTIONS

- **Terrain-Ready:** Targets LA's fire-prone hills with summit sprinklers and drone monitoring.
- **Smart Water Use:** Uses stormwater and deep groundwater, easing hydrant demand.
- **Fast Detection:** Satellite-linked AI spots and responds to fires instantly.
- **Scalable Design:** Grows from neighborhood to city-wide, using sustainable water.
- **Fair Access:** Brings fire protection to underserved and remote communities.



7

SMOKE POLLUTION RESPONSE

- Attached to building exterior to autonomously function as smoke levels surpass a threshold, purifying the air